OPTIMIZATION OF EMERGENCY EVACUATION ROUTES IN CALUMPIT USING NETWORK FLOW MODELS AND LINEAR PROGRAMMING

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ABSTRACT

This study analyzes emergency evacuation routes in Barangay Corazon, Calumpit using graph theory, network flow models, and linear programming. The evacuation system is modeled as a network where Barangay Corazon serves as the source, Barangay Poblacion acts as a transit route, and the evacuation center is the sink. With a population of 4,556 residents and limited transportation consisting of one ambulance and one tricycle patrol, the model determines a maximum evacuation flow of 16 to 19 persons per cycle, resulting in 240 to 285 required trips and an evacuation time of approximately 32 to 38 hours under knee-level flooding conditions. The study highlights how transportation limitations and flood conditions affect evacuation efficiency.

Keywords: graph theory, network flow, linear programming, evacuation planning, flood evacuation, optimization

THE PROBLEM AND ITS BACKGROUND

Flooding is one of the most frequent natural hazards in the Philippines due to the country's exposure to heavy and prolonged rainfall caused by weather phenomena such as monsoons, thunderstorms, and tropical cyclones (PAGASA, n.d.). These conditions often lead to rising river levels and the accumulation of rainwater in low-lying areas. When rivers and drainage systems exceed their capacity, water overflows and floods nearby communities. Calumpit in particular is highly vulnerable due to its low-lying geography and proximity to major rivers such as the Pampanga River and the Angat River, which cause water to accumulate in surrounding areas during heavy rainfall or typhoon events.

Floods in Calumpit have extensive effects on the community. Homes, roads, bridges, and public buildings may be submerged in floodwater, limiting mobility and access to essential services. Agricultural crops and livestock are also at risk, while commercial and industrial activities can be disrupted, leading to economic losses. Floodwaters may displace residents, create shortages of food and clean water, and increase the risk of waterborne diseases.

In high-risk areas such as Barangay Corazon, floodwaters can rise rapidly, making timely evacuation extremely important. Residents must be transported to designated evacuation centers before water levels become too dangerous for travel. However, evacuation operations are often challenged by limited rescue vehicles, narrow road networks, and the need to move a large number of residents within a short period of time. In many situations, evacuees from Barangay Corazon must pass through nearby routes in Barangay Poblacion, which serves as a transit path leading to the evacuation center.

Another factor that affects evacuation efficiency is the presence of floodwater during an emergency. Even at knee-level flooding, vehicle movement becomes slower and more difficult, reducing the efficiency of rescue operations and potentially delaying evacuation efforts (Rua Seguridad, 2023).

To address these challenges, mathematical approaches such as network flow models and linear programming can be applied to evacuation planning. These methods allow road networks to be represented as mathematical graphs, where locations are treated as nodes and roads as edges. By analyzing how evacuees move through these networks, it becomes possible to identify bottlenecks, evaluate transportation capacity, and determine more efficient evacuation strategies.

This study applies mathematical optimization techniques to analyze emergency evacuation routes for residents of Barangay Corazon in Calumpit, Bulacan. The evacuation network is modeled using graph theory where Barangay Corazon acts as the source node, the routes passing through Barangay Poblacion serve as a transit path, and the evacuation center functions as the sink node. By incorporating transportation capacity, road limitations, and flood conditions, the model aims to determine strategies that can improve evacuation efficiency and reduce evacuation time during flood events.

Statement of the problem and Objective of the study

The central challenge of this research lies in the difficulty of evacuating residents from flood-prone areas of Barangay Corazon during severe flooding events. This study primarily addresses the issue of infrastructure limitations, particularly how narrow streets and limited transportation resources can restrict the movement of evacuees toward safe evacuation centers. (Bounce Readiness, 2025)

Specifically, this research examines how a limited number of rescue vehicles can be efficiently utilized to transport residents from Barangay Corazon to designated evacuation centers through routes that pass through Barangay Poblacion, which serves as a transit route during evacuation operations. As floodwaters rise, transportation becomes more difficult, further slowing evacuation efforts and increasing the risk to residents.

Another challenge considered in this study is the effect of flood conditions on evacuation efficiency. Even at knee-level flooding, vehicle movement becomes slower and

more difficult, which may significantly delay evacuation operations and reduce overall system efficiency.

Given these conditions, it becomes necessary to determine how evacuation routes and transportation resources can be optimized to move residents safely and efficiently. Therefore, this study investigates how mathematical optimization techniques, particularly network flow models and linear programming, can be applied to identify efficient evacuation strategies and possible bottlenecks within the evacuation network.

General Objective

The primary goal of this research is to determine the most efficient evacuation strategy for the residents of Barangay Corazon using mathematical optimization.

Specific Objectives

1. To determine the evacuation demand of Barangay Corazon residents who must be transported to designated evacuation centers during flooding events.
2. To evaluate how the limited number of available rescue vehicles can transport evacuees under knee-level flood conditions.
3. To assess how knee-level flooding affects travel time and overall evacuation efficiency.
4. To analyze evacuation routes passing through Barangay Poblacion as a transit path and identify possible road segments that may cause congestion or delay.

Scope and Limitation

This study focuses on the optimization of emergency evacuation routes in Barangay Corazon, Calumpit, Bulacan using graph theory, network flow models, and linear programming. The evacuation system is modeled as a directed weighted graph where intersections are represented as nodes and roads are represented as edges. The study considers only knee-level flooding conditions, assuming that all identified roads remain passable but with reduced travel speed.

Barangay Corazon is treated as the source of evacuees, while Barangay Poblacion is considered only as a transit route leading to the evacuation center. The transportation system is limited to one patrol ambulance and one tricycle patrol with fixed capacity ranges. The model applies mathematical optimization techniques to maximize evacuation flow and minimize evacuation time under the given constraints.

Scope of the Study

1. Focuses only on Barangay Corazon evacuation system
2. Considers knee-level flooding conditions only
3. Models the road network using graph theory
4. Uses one ambulance and one tricycle as transport units
5. Applies network flow and linear programming techniques

Limitations of the Study

The study is subject to several limitations. It assumes constant vehicle availability and does not consider breakdowns or operational delays. Human behavior, panic, and decision-making during evacuation are not included in the model. Additionally, traffic conditions, road obstructions, and weather variations beyond knee-level flooding are not considered. The study also assumes fixed travel speed and does not account for real-time changes in road conditions.

1. Assumes constant vehicle availability
2. Does not include human behavior or panic situations
3. Ignores traffic congestion and road obstructions
4. Assumes fixed travel speed
5. Does not consider dynamic or real-time changes

Significance of the Study

This study is significant in both mathematical and practical contexts. From a mathematical perspective, it demonstrates the application of graph theory, network flow

models, and linear programming in solving real-world problems. It highlights the use of the Max-Flow Min-Cut Theorem in analyzing evacuation systems and identifying bottlenecks within a network.

From a practical standpoint, the study provides insights into evacuation challenges in flood-prone communities such as Barangay Corazon. By identifying limitations in transportation capacity and route efficiency, the results can help local planners and decision-makers improve evacuation strategies and resource allocation. Furthermore, this study contributes to academic learning by serving as a reference for students studying graph theory and its applications.

1. Demonstrates real-life application of graph theory and network flow
2. Identifies bottlenecks in evacuation systems
3. Helps improve evacuation planning and decision-making
4. Provides insight for flood-prone community management
5. Serves as an academic reference for graph theory applications

GRAPH-THEORETIC FOUNDATIONS OF THE MODEL

Review of Related Literature

Several researchers have explored how mathematical models can help improve emergency evacuation planning. In an earlier study, Yamada (1996) used a network flow approach to model a city during an evacuation. In the study, different locations were treated as nodes while roads were considered edges in a graph. The goal was to assign residents to safe evacuation centers using the shortest available routes. The model also took into account the capacity limits of evacuation centers by transforming the problem into a minimal cost flow model. The results showed that network flow methods can help planners analyze evacuation systems and improve the efficiency of evacuation routes.

Another study by Hong et al. (2018) focused on evacuation planning in restricted spaces where congestion is more likely to occur. They introduced the Congestion-Avoidable Evacuation Route Network Planning (CA-ERNP) problem and proposed an algorithm based on the minimum-cost maximum-flow model. Their approach aimed to reduce congestion and shorten evacuation time. Based on their simulation experiments, the algorithm performed better than some existing methods and helped improve the overall evacuation process.

Similarly, Lee (2014) developed a route optimization model designed for emergency evacuation and disaster response. Their research combined several optimization techniques such as the Dijkstra shortest path algorithm, genetic algorithms, and Pareto efficiency methods. The model was tested using both a virtual network and an actual transportation network in the Gumi National Industrial Complex in Korea. The results indicated that the optimized routes were more efficient compared to routes that were not optimized.

These studies show that mathematical optimization and network flow techniques can play an important role in improving evacuation planning and reducing congestion

during disasters. Building on these ideas, the present study applies the Max-Flow Min-Cut framework and linear programming to optimize emergency evacuation routes in Barangay Corazon, Calumpit, Bulacan. The goal is to minimize evacuation time and identify possible road bottlenecks that may occur during flooding events.

Theoretical Framework

This study is held on mathematical and optimization theories that explain how evacuation systems can be modeled and improved during disaster situations. Specifically, the research is based on Graph Theory, Network Flow Theory, Linear Programming, and the Max-Flow Min-Cut Theorem developed by (Ford & Fulkerson, 1956). These theories provide the foundation for modeling evacuation routes and optimizing the movement of evacuees during flooding in Barangay Corazon, Calumpit, Bulacan.

Graph Theory provides a mathematical structure for representing networks consisting of vertices and edges. In this study, residential areas, road intersections, and evacuation centers are represented as vertices, while the streets connecting them are represented as edges. This framework allows the road network of Barangay Corazon to be analyzed as a mathematical graph, making it possible to study connectivity and determine efficient evacuation routes. Graph-based representations are widely used in transportation and logistics optimization problems (Ahuja et al., 1993).

The study also applies Network Flow Theory, which examines how resources move through a network with limited capacities. In the evacuation model, evacuees represent the flow moving through the network from unsafe areas to safe evacuation centers. Each road segment has a capacity that represents the maximum number of evacuees or vehicles that can pass through within a given period. This concept helps analyze congestion and determine how evacuees can be distributed efficiently across available routes.

A central concept in this research is the Max-Flow Min-Cut Theorem, which states that the maximum possible flow in a network is equal to the capacity of the minimum

cut that separates the source from the sink. In the context of evacuation planning, this theorem helps identify critical road segments or intersections that act as obstacles and limit the movement of evacuees. Identifying these obstacles allows planners to improve evacuation routes and minimize delays during emergencies.

In addition, the study applies Linear Programming to determine the optimal evacuation strategy. Linear programming is used to maximize the number of evacuees transported from residential areas to evacuation centers while considering several constraints such as road capacity, flood accessibility, rescue vehicle availability, and evacuation center capacity. Through this framework, the evacuation problem can be analyzed mathematically to determine optimal routes, identify obstacles, and estimate the minimum time required to safely transport residents to evacuation centers during flooding events.

Objective Formulation

To remove ambiguity, this study adopts a maximum flow objective. The goal is to maximize the number of evacuees transported per unit time through the evacuation network.

Maximize $f(s,t)$

subject to:

$0 \leq f(u,v) \leq c(u,v)$

$\sum f_{in} = \sum f_{out}$

where:

$f(s,t)$ is the total evacuation flow

$c(u,v)$ is edge capacity

Evacuation time is derived from flow:

$Time = \frac{Population}{Flow \times T_{trip}}$

Max-Flow Min-Cut Theorem

The Max-Flow Min-Cut Theorem states that the maximum possible flow from a source node s to a sink node t in a flow network is equal to the total capacity of the minimum cut that separates s from t.

$\max f(s,t) = \min \sum_{u \in S, v \in T} c(u,v)$

where:

$f(s,t)$ represents the total flow from the source node s to the sink node t

$c(u,v)$ represents the capacity of the edge from node u to node v

(S,T) is a partition of the vertex set such that $s \in S$ and $t \in T$

In evacuation modeling, the source node represents flood-prone residential areas, while the sink node represents the evacuation center. The theorem helps identify bottleneck road segments whose limited capacity restricts the total number of evacuees that can move through the network.

METHODOLOGY

This section presents the process of modeling the evacuation system of Barangay Corazon using graph theory, network flow models, and linear programming. Graph-based evacuation models are effective in simulating real-world disaster scenarios and evaluating system performance (Campos et al., 2012). It describes how real-world data is translated into a mathematical model and how optimization techniques are applied to determine efficient evacuation strategies.

Description of the Real-Life System

This study focuses on Barangay Corazon in Calumpit, Bulacan. Only the residents of Barangay Corazon are included in the evacuation demand, while the streets in Barangay Poblacion are treated solely as transit routes leading to the evacuation center and are assumed to remain passable during flooding.

The total number of residents in Barangay Corazon is:

$$P = 4{,}556$$

The priority evacuation area during flooding is Purok 1 (De Jesus Street and M. Serrano Street), consisting of:

- Purok 1A = 193 families
- Purok 1B = 117 families
- Purok 1C = 118 families

$$\text{Total}_{\text{Purok 1}} = 428 \text{ families}$$

Using the estimated average household size of 4.88 persons, where:

$$\text{f} = \text{Number of families}$$

$$\text{I} = \text{Average household size}$$

$$\text{Residents}_{\text{Purok 1}} = f \times I = \text{ Number of Individuals}$$

$$\text{Residents}_{\text{Purok 1}} = 428 \times 4.88 = 2{,}089 \text{ persons}$$

The available rescue vehicles are:

- 1 Patrol Ambulance (12–14 persons)
- 1 Tricycle Patrol (4–5 persons)

Where:

$$C_{A1} = \text{Minimum Capacity of Ambulance}$$

$$C_{A2} = \text{Maximum Capacity of Ambulance}$$

$$C_{T1} = \text{Minimum Capacity of Tricycle Patrol}$$

$$C_{T2} = \text{Maximum Capacity of Tricycle Patrol}$$

Minimum transport capacity:

$$C_{\min} = C_{A1} + C_{T1}$$

$$C_{\min} = 12 + 4 = 16 \text{ persons}$$

Maximum transport capacity:

$$C_{\max} = C_{A2} + C_{T2}$$

$$C_{\max} = 14 + 5 = 19 \text{ persons}$$

The total population was obtained from barangay records. The number of residents in Purok 1 was computed by multiplying the number of families (428) by the average household size of 4.88 persons. The transport capacity was determined by adding the capacities of the available rescue vehicles.

Network Representation and Flow Optimization

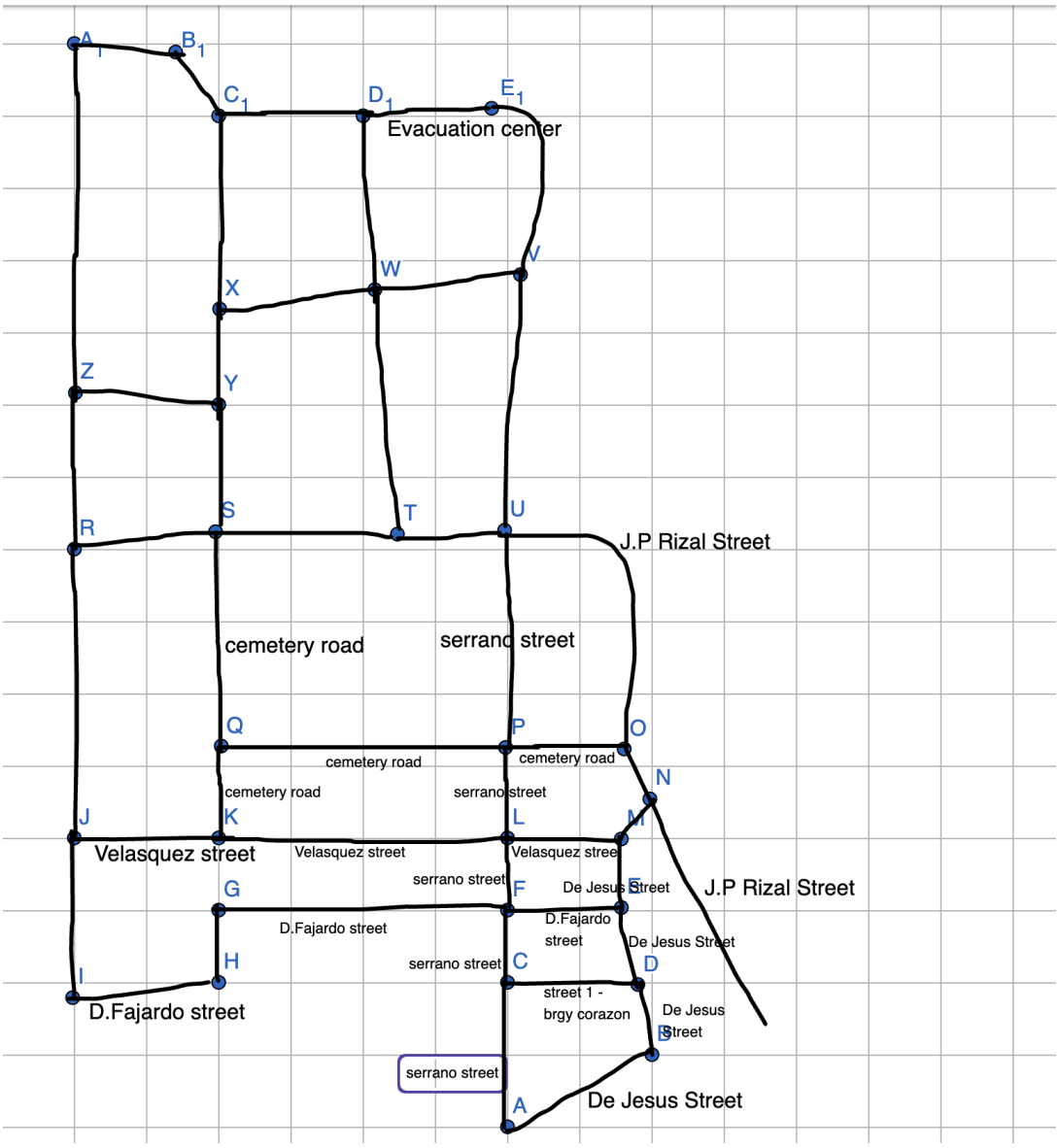

Figure 1

Street Network Graph of Barangay Corazon with Labeled Nodes

Table 1

Measured Road Segment Lengths in Barangay Corazon

Edge	Length	Edge	Length	Edge	Length
A–B	130.83	A–C	122.30	B–D	47.12
C–D	91.45	C–F	44.71	D–E	49.12
E–F	73.65	E–M	79.84	F–G	117.83
F–L	84.51	G–H	47.27	H–I	106.17
I–J	155.85	J–K	93.24	J–R	183.92
K–L	136.71	K–Q	71.92	L–M	70.90
L–P	78.01	M–N	34.15	N–O	60.92
O–P	91.97	O–U	254.80	P–Q	133.85
P–U	180.46	Q–S	161.62	R–S	102.33
R–Z	123.45	S–Y	77.74	S–T	79.05
T–U	57.62	T–W	138.11	U–V	162.40
V–W	91.66	V–E_1	99.16	W–X	81.55
W–D_1	105.41	X–Y	53.91	X–C_1	115.45
Y–Z	90.63	Z–A_1	202.64	A_1–B_1	55.26
B_1–C_1	35.30	C_1–D_1	80.03	D_1–F_1	51.74
E_1–F_1	29.40				

The evacuation network is modeled as a directed weighted graph:

$$G = (V, E)$$

Ground elevation ranking from lowest to highest is:

1. Serrano Street (Nodes A–P)
2. De Jesus Street
3. Street 1
4. D. Fajardo Street
5. Velasquez Street
6. Cemetery Road (K–S, Q–P)
7. Serrano Street (P–U)
8. Cemetery Road (P–O)
9. J.P. Rizal Street

Evacuation begins at the lowest elevation (Node A area). The shortest path from

the lowest node (A) to the evacuation center (F_1) is:

$$A \to B \to D \to E \to M \to N \to O \to U \to V \to E_1 \to F_1$$

Total measured distance:

$$947.74 \text{ meters}$$

$$15 \text{ km/h} = 4.17 \text{ m/s}$$

$$t = \frac{947.74}{4.17} = 227.2 \text{ seconds}$$

$$t = 3.79 \text{ minutes}$$

$$T_{\text{trip}} = 2(3.79) = 7.58 \text{ minutes}$$

$$T_{\text{trip}} \approx 8 \text{ minutes per evacuation cycle}$$

Modelling Assumptions and Constraints

The model assumes that all roads remain passable during knee-level flooding, with reduced travel speed. Rescue vehicles are assumed to operate continuously without delay. The system uses fixed vehicle capacity and assumes constant travel time. Constraints include limited transportation resources, road accessibility, and evacuation demand.

Applied Graph-Theoretic Algorithms

Max-Flow Model

The system is modeled as a flow network to maximize evacuation capacity.

Ford-Fulkerson Method

The Ford-Fulkerson method is used to compute the maximum flow in the evacuation network by identifying augmenting paths from the source node to the sink node.

The identified shortest path A → B → D → E → M → N → O → U → V → E1 → F1 serves as the primary augmenting path in the evacuation network. Following the Ford-Fulkerson approach, flow is pushed from the source node to the sink node through this path while respecting system capacity constraints.

In this case, the limiting factor is not the road network but the available rescue vehicles. The patrol ambulance can transport 12–14 persons, while the tricycle patrol can transport 4–5 persons, resulting in a total transport capacity of 16 to 19 persons per evacuation cycle.

Since all evacuees rely on the same transportation system, no additional augmenting path can increase the total flow without increasing vehicle capacity. Therefore, the system reaches its maximum flow after this augmentation, as no further increase in flow is possible under the given constraints.

Shortest Path Analysis

The shortest path analysis is used to determine the most efficient route from the source node (Barangay Corazon) to the evacuation center. Using the weighted graph representation of the road network, the path

A → B → D → E → M → N → O → U → V → E1 → F1

was identified as the shortest route with a total distance of 947.74 meters.

This shortest path minimizes travel time under normal conditions. However, during flooding, reduced vehicle speed increases travel time, which directly affects evacuation efficiency. The computed travel time along this path is approximately 3.79 minutes per trip and about 8 minutes per round-trip cycle.

This analysis shows that while the route itself is optimal in terms of distance, overall evacuation performance is still constrained by transportation capacity rather than path efficiency.

RESULTS AND DISCUSSION

This section presents the computed results of the evacuation model and provides both mathematical and real-world interpretations. The analysis focuses on identifying system limitations and evaluating the efficiency of the evacuation process.

Evacuation Demand

The total population of Barangay Corazon is given by

$$P = 4,556.$$

Among these, approximately 2,089 residents belong to priority areas that are most vulnerable to flooding. This indicates that a large portion of the population requires immediate evacuation during disaster events.

Interpretation: The high evacuation demand highlights the need for an efficient and well-structured evacuation system. A large number of evacuees increases pressure on transportation resources and evacuation routes.

Transportation Capacity

The available rescue vehicles consist of one ambulance and one tricycle patrol. Their combined transport capacity per trip is:

$$16 \leq f_{\max} \leq 19.$$

Interpretation: This limited transport capacity represents a critical constraint in the evacuation system. Regardless of how efficient the routes are, the number of evacuees transported per cycle remains restricted by vehicle capacity.

Evacuation Trips Required

The number of evacuation trips required is computed as:

$$N = \frac{4,556}{f_{\max}},$$

resulting in approximately

$$240 \text{ to } 285 \text{ trips.}$$

Interpretation: The large number of required trips indicates that the evacuation system is highly dependent on repeated transport cycles. This reveals inefficiency in the system, especially under time-sensitive emergency conditions.

Travel Time Analysis

The estimated round-trip time per evacuation cycle is:

$$T_{\text{trip}} \approx 8 \text{ minutes.}$$

Thus, the total evacuation time is:

$$240 \times 8 = 32 \text{ hours}, \quad 285 \times 8 = 38 \text{ hours.}$$

Interpretation: An evacuation duration of 32 to 38 hours is impractical during emergency flooding scenarios. This indicates that the current system cannot support rapid evacuation, which is critical for minimizing risk and casualties.

Maximum Flow Interpretation

The maximum flow of the evacuation network is:

$$16 \leq f_{\max} \leq 19.$$

Interpretation: Based on the Max-Flow Min-Cut Theorem, the transportation system acts as the bottleneck of the network. The limitation is not due to the road structure but rather the capacity of available rescue vehicles.

Scenario Analysis

If one additional tricycle is introduced, the transport capacity increases to:

$$20 \leq f_{\max} \leq 24.$$

This reduces the total evacuation time to approximately:

$$25 \text{ to } 30 \text{ hours}.$$

Interpretation: The results demonstrate that increasing transportation capacity significantly improves evacuation efficiency. This suggests that resource allocation, rather than route optimization alone, plays a crucial role in enhancing evacuation performance.

SUMMARY AND CONCLUSION

Summary of Findings

This study modeled the evacuation system of Barangay Corazon using graph theory, network flow models, and linear programming. The results show that while the road network provides feasible evacuation routes, the system is significantly constrained by limited transportation capacity.

With a total population of 4,556 residents and only one ambulance and one tricycle patrol, the system can transport only 16 to 19 individuals per cycle. This results in approximately 240 to 285 evacuation trips and a total evacuation time ranging from 32 to 38 hours.

Conclusions

The findings confirm that the primary limitation of the evacuation system lies in transportation capacity rather than network structure. Even with optimized routes, the system remains inefficient due to limited vehicle availability.

From a graph-theoretic perspective, the transportation capacity represents the minimum cut of the network, restricting the maximum flow. Therefore, increasing the number of available rescue vehicles is the most effective way to improve evacuation efficiency.

This study demonstrates that mathematical modeling can provide valuable insights into real-world disaster management systems. Future improvements may include incorporating dynamic traffic conditions, multiple evacuation centers, and real-time optimization models to enhance system performance.

These findings emphasize that in real-world evacuation planning, improving infrastructure alone is insufficient. Instead, strategic allocation of transportation

resources plays a more significant role in increasing evacuation efficiency. This highlights the importance of integrating mathematical modeling with practical disaster management policies.

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APPENDIX A

Edge List of the Evacuation Network

This appendix presents the complete list of road segments and their corresponding lengths used in constructing the evacuation network model of Barangay Corazon.

Table 2

Measured Road Segment Lengths in Barangay Corazon

Edge	Length	Edge	Length
A–B	130.83	A–C	122.30
B–D	47.12	C–D	91.45
C–F	44.71	D–E	49.12
E–F	73.65	E–M	79.84
F–G	117.83	F–L	84.51
G–H	47.27	H–I	106.17
I–J	155.85	J–K	93.24
J–R	183.92	K–L	136.71
K–Q	71.92	L–M	70.90
L–P	78.01	M–N	34.15
N–O	60.92	O–P	91.97
O–U	254.80	P–Q	133.85
P–U	180.46	Q–S	161.62
R–S	102.33	R–Z	123.45
S–Y	77.74	S–T	79.05
T–U	57.62	T–W	138.11
U–V	162.40	V–W	91.66
V–E1	99.16	W–X	81.55
W–D1	105.41	X–Y	53.91
X–C1	115.45	Y–Z	90.63
Z–A1	202.64	A1–B1	55.26
B1–C1	35.30	C1–D1	80.03
D1–F1	51.74	E1–F1	29.40

APPENDIX B

Shortest Path Computation Details

The shortest path from the source node (A) to the evacuation center (F1) is given by:

$$A \to B \to D \to E \to M \to N \to O \to U \to V \to E1 \to F1$$

The total distance is:

$$947.74 \text{ meters}$$

Travel time is computed as:

$$t = \frac{d}{v} = \frac{947.74}{4.17} = 227.2 \text{ seconds}$$

$$t \approx 3.79 \text{ minutes}$$

Round-trip time:

$$T_{\text{trip}} \approx 8 \text{ minutes}$$